

Automated Redox Titrations via Interdigitated Electrode Arrays: Application to the Mediated Electron Transfer Interrogation of Charge and Rate on Electrodeposited Polymers

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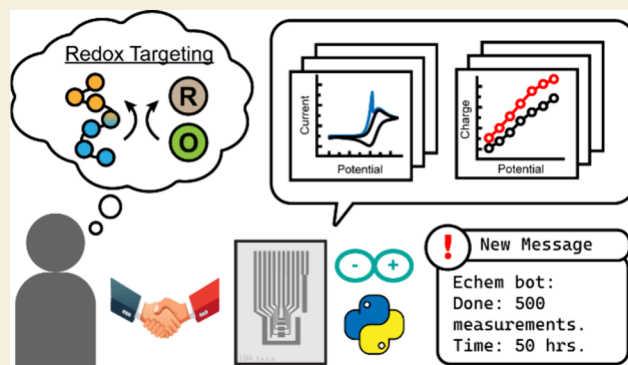
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ABSTRACT: Mediated electron transfer (MET) plays a crucial role in energy storage and conversion technologies such as redox targeting flow batteries (RTFBs), yet its experimental investigation often requires labor-intensive and low-throughput setups. To address this, we developed a microfabricated interdigitated electrode array (IDA) platform that enables automated, high-throughput electrochemical redox titration measurement to be performed to study the MET process. Our redox titration method enables simultaneous measurement of the charge capacity and rate of MET processes on a material or surface. Automated redox titration (ART) facilitates systematic investigation of the MET process across a broad parameter space, exemplified through the study of polypyrrole (PPy) and a pyrene-4,5,9,10-tetrone azo group-based polymer (PTAP), both redox-active polymers relevant to various energy storage applications. Using PPy as a model material, 500 redox titration measurements were conducted within 50 h, varying the electrode gap widths, polymer charging potentials, voltammetric scan rates, and electrolyte concentrations. Finite-element simulations confirmed the electrochemical responses and elucidated the kinetics of the MET reactions. Our automated methodology was further tested with PTAP, revealing a surprising charging potential dependence on the rate of MET. The automation, flexibility, and scalability of our redox titration platform pave the way not only for advanced studies of MET processes relevant to RTFBs, but also with implications in the understanding of next-generation energy storage materials, molecular electrocatalysis, and biosensing.

KEYWORDS: Automation, microfabrication, redox titration, interdigitated electrode arrays, redox targeting, redox-active polymers



INTRODUCTION

Mediated electron transfer (MET) is essential to several chemical processes that range from cellular respiration to chemical synthesis. In electrochemical processes, MET is often used as an alternative way of carrying out reactions that are inefficient through direct electron transfer.^{1–4} In general, MET, when used in an electrochemical setup, involves a heterogeneous electron transfer to a freely diffusing redox mediator, which can then undergo a subsequent electron transfer step with a substrate material (Figure 1). MET in the solution phase has found uses in various applications such as electrochemical sensing of biomarkers like glucose,^{5,6} molecular electrocatalysis,^{7–9} and mediated electrosynthesis.^{10,11} While there are many examples of MET in the solution phase, redox mediators are also used to transfer charge to and from solid substrates, in a process reminiscent of a redox titration of the substrate.

The use of redox mediators to transfer charge between two stationary species is crucial to emerging energy storage and

conversion processes. Photovoltaic devices, such as dye-sensitized solar cells, use iodide as a redox mediator to shuttle charges between electrodes.¹² Next generation batteries, such as lithium–air batteries, rely on redox mediators to improve efficiency and lifetime of the battery.¹³ Recently, MET-based approaches have been used to improve the energy density of redox-flow batteries (RFBs). RFBs are one of the most promising emerging technologies for grid-level energy storage.^{14–17} RFBs exhibit decoupled power generation and energy storage,¹⁷ high capacity,¹⁸ and excellent operational safety.¹⁹ However, they typically face challenges such as high costs and poor energy density, primarily due to the limited

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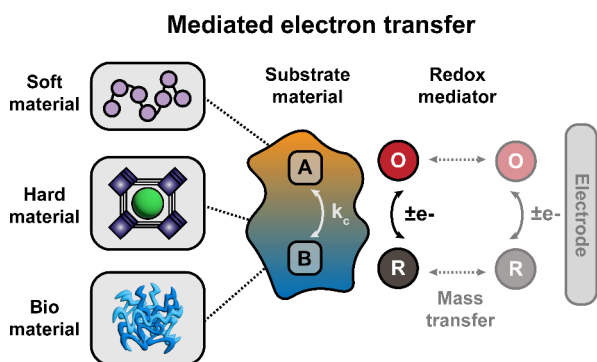


Figure 1. Schematic illustration of the mediated electron transfer (MET) process. k_c is the rate constant of the reaction between the redox mediator and the substrate material. In the presence of a reactive mediator, this reaction proceeds until charges in the substrate materials have been fully consumed.

solubility of redox-active species in the electrolyte solvents.^{20–22} To circumvent these issues, the redox targeting flow battery (RTFB) has been introduced, using a solid-state energy storage material to achieve effective charge storage concentrations up to ~ 20 M.^{23–27} RTFBs operate through MET between a freely diffusing redox mediator and the solid energy storage material, effectively titrating the latter via the former. This process occurs through a redox-targeting reaction that is driven by differences in Nernstian potentials that renders a spontaneous chemical reaction between the mediator and the energy storage material.²⁸ An ideal RTFB system should offer high charge storage capacity while maintaining fast kinetics of the redox-targeting reaction.²⁹

Optimizing the interaction between the redox mediator and solid energy storage material requires searching across a wide range of redox mediators, energy storage materials, and electrolyte conditions.^{30–32} Despite recent efforts to model the MET processes in RTFBs, there still lacks versatile analytical methods that can accelerate experimental inves-

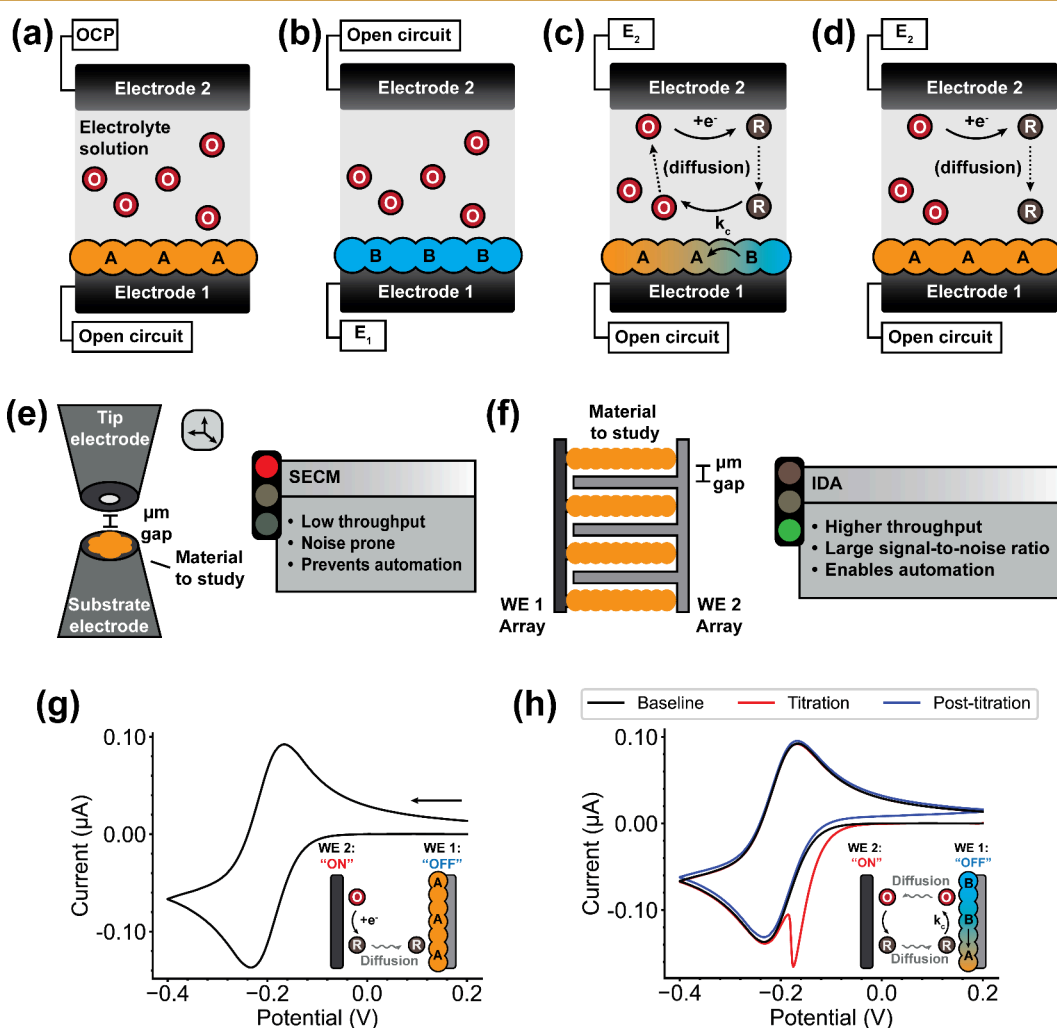
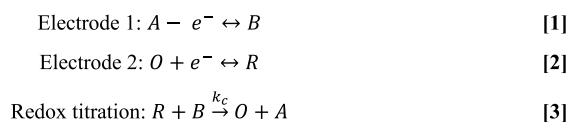


Figure 2. Principle of redox titration measurements with SECM and IDA. (a) Initial condition of a redox titration measurement. (b) Charging of the energy storage material on electrode 1 while electrode 2 is at open circuit. (c) Redox titration of the charged energy storage material with electrode 2 while electrode 1 is at open circuit. (d) Completion of the titration process for the energy storage material. (e) The SI-SECM setup and its drawback. (f) The setup of redox titration measurement with IDA and its advantages. (g) Simulated cyclic voltammetry (CV) response in the absence of the titration reaction on an IDA. The black arrow indicates the initial scan direction of the CV. (h) CV response in the presence of the titration reaction. The post-titration CV overlaps with the baseline, indicating the completion of the titration reaction. Inset schematics in (g) and (h) indicate the electrochemical and chemical processes happening during the simulated CV measurements.

tigation of redox-targeting reactions across a broad parameter space.³³ RTFBs are typically investigated through charge–discharge cycling of the full battery.^{24,34,35} However, this is time-consuming, which typically takes several hours to complete a charge–discharge cycle, depending on the current density applied. Moreover, RTFB cycling measurements provide limited mechanistic insight into the redox-targeting reaction. Our group has previously used scanning electrochemical microscopy (SECM) to perform microscale redox-targeting reactions, probing the dynamics of charge transfer within redox active colloid energy storage materials.³⁶ These microscale redox-targeting experiments were an extension of the surface interrogation (SI) technique, an SECM method that uses redox mediators to titrate redox-active surface species.^{36–38} These methods are highly relevant to RTFB studies, because it has been recently discussed that the kinetics of the MET process determine the efficiency of the redox targeting process.^{31,34} Thus, developing new methods with high throughput to understand MET reactions is greatly needed.

Generally, the SI experiment involves a bipotentiostat setup with two working electrodes in close proximity, allowing their diffusion layers to overlap. Electrode 1, typically a substrate electrode such as a catalyst or battery material, generates a redox-active species that is confined to its surface. Electrode 2, typically an SECM tip in SI-SECM measurements, generates the reactive form of a freely diffusing redox mediator that can titrate the species on electrode 1. Below we consider the SI of an energy storage material (A/B) deposited on electrode 1 and can be oxidatively charged (i.e., generating species B) while in the presence of a freely diffusing redox mediator (O/R) (Figure 2a). First, an oxidative potential, E_1 , is applied to electrode 1 to charge the material, producing its B form, while electrode 2 is left at open circuit (Figure 2b). Then, with electrode 1 at open circuit, a reducing potential, E_2 , is applied to electrode 2 to reduce the redox mediator, which in its R form diffuses to electrode 1 to titrate species B. The titration reaction proceeds with a rate constant k_c , regenerating the original form of the redox mediator (O) and of the charge storage material (A). The entire process is summarized in Scheme 1.

Scheme 1. General Steps in an SI Measurement



The final step of Scheme 1 gives rise to an increased flux of O as the reaction proceeds. In SI-SECM this is observed as a positive feedback loop that transiently increases the current measured by electrode 2 (Figure 2c). The positive feedback from the titration reaction stops when the charged energy storage material, B, has been fully consumed (Figure 2d). While SI-SECM is a powerful technique, it has limitations that make it unsuitable for high-throughput or automated analysis. These limitations include complicated manual operations such as approaching and coaligning two similarly sized ultramicroelectrodes with a piezo positioner (Figure 2e).³⁹ Additionally, the small current output by ultramicroelectrodes (typical ranges from pA to nA) leads to small redox titration current in SI-SECM measurements, which is highly susceptible

to mechanical and electrical interferences,⁴⁰ making reproducible and quantitative analysis cumbersome. A microfluidic method introduced by Anderson and Crooks performed SI-like measurements in a generation-collection geometry that used band electrodes inside a microscale channel, showing that SI-SECM-type techniques could be scaled to analyze samples at a larger scale.⁴¹ However, irreversible confinement of the electrodes inside the channel prevents ex-situ analysis and surface modifications.

Microfabricated interdigitated electrode arrays (IDA) offer an automation-friendly way to carry out redox titration measurements, offering enhanced throughput and signal-to-noise ratios (Figure 2f).^{42,43} IDA devices consist of two independently controlled working electrode (WE) arrays with micron-scale gap widths (Figure 2f). Redox titration measurements performed using IDAs function almost identically to those performed with SI-SECM. Simulated voltammograms show the expected response of a redox titration measurement with IDAs (Figure 2g), with a characteristic peak resulting from transient positive feedback. Following this peak, the voltammogram retains its original shape upon the completion of the titration reaction (Figure 2h). This shape is analogous to the total catalysis case described in molecular electrocatalysis theory.¹ More details about the simulations are available in the Experimental Section.

In this work, we demonstrate the capability of performing automated redox titration (ART) measurements with IDAs to study MET for energy storage materials. We microfabricated IDA devices that contain four different IDA gap widths to mimic different tip–substrate distances in SI-SECM measurements. We then applied the ART methodology to study charge storage on polypyrrole (PPy) as a model system. PPy is a conducting polymer that has found use in electrochemical supercapacitors and batteries.^{44–47} We started by selectively electrodepositing PPy thin film onto one of the electrode arrays of the IDA. We then performed ART measurements on PPy under aqueous media with a range of different PPy charging potentials and CV scan rates. Using our Hard Potato Python library,⁴⁸ we controlled potentiostat functions in an automated fashion, enabling swift exploration of titration parameters, and dramatically improving the throughput of redox titration processes compared to SI-SECM. Redox titration measurements with elevated supporting electrolyte concentration illustrated the effect of ionic strength on the charge storage behavior of PPy. We performed over 500 CVs to systematically explore this system, demonstrating the needed versatility and reliability of the automated redox titration methodology for advanced studies of energy storage materials.

Finally, to demonstrate the generalizability of our ART methodology to emerging materials, we highlight the charge storage behavior of pyrene-4,5,9,10-tetrone azo group-based polymer (PTAP), a recently developed pyrene-4,5,9,10-tetraone (PTO) derivative.⁴⁹ PTAP is a conjugated, multi-electron redox-active material used in metal-ion battery cathodes,⁴⁹ which we believe could be used as a model electropolymerizable PTO derivative for RTFB studies. We electropolymerized PTAP from its monomer form, 2,7-dinitropyrene-4,5,9,10-tetraone (PTO-2NO₂) and performed automated redox titration of PTAP using ferrocene as the redox mediator. Since PTAP possesses multiple redox-active moieties, we explored a series of charging potentials to fully access the redox-active units. We observed the ability for

ferrocene to titrate the charge stored in PTAP greatly depends on the applied charging potential, highlighting the multi-electron nature of the polymer and pointing out future directions to better understand the charge transfer process between redox mediators and the redox-active units in multielectron redox-active polymers.

■ EXPERIMENTAL SECTION

Chemicals and Materials

18-M Ω DI water was obtained from a Mill-Q purification system. Pyrrole (reagent grade, 98%), hydroxymethylferrocene (FcMeOH, 97%), hexaammineruthenium(III) (Ruhex) chloride (98%), potassium phosphate monobasic ($\geq 99\%$), and potassium phosphate dibasic (reagent grade, $\geq 98\%$), ferrocene (Fc, 98%), anhydrous lithium perchlorate (LiClO₄), and anhydrous acetonitrile (MeCN) were purchased from Sigma-Aldrich (St. Louis, MO). Potassium nitrate ($\geq 99\%$) was purchased from VWR chemicals (Radnor, PA). Potassium chloride (KCl, USP grade) was purchased from Research Products International (Mount Prospect, IL). Tetrabutylammonium hexafluorophosphate (TBAPF₆, $>98\%$) was purchased from TCI America (Portland, OR). MeCN (HPLC grade) and sulfuric acid (H₂SO₄, trace metal grade) were purchased from Thermo Fisher Scientific (Waltham, WA). Fc and LiClO₄ were vacuum-dried using Schlenk flasks overnight before transferring into the glovebox. Other chemicals were used as received. 4-in. glass wafers used to fabricate the IDA devices were purchased from University Wafer (South Boston, MA). The photoresists, developers, and solvents used in the fabrication process were purchased from Kayaku Advanced Materials (Westborough, MA). The flexible cables and anisotropic conductive tape used to render electrical connection to the IDAs were purchased from DigiKey Electronics (Thief River Falls, MN) and 3 M (Saint Paul, MN), respectively.

Device Fabrication and Packaging

The IDAs were fabricated on 4-in. glass wafers following a previously reported workflow (Figure S1, microfabrication pathway of the IDA device).^{42,43} Initially, the wafer was degreased with acetone, isopropanol, and DI water, followed by O₂-plasma and RCA-1 cleaning. Then, an SPR220–4.5/LOR 5A bilayer photoresist was spun onto the wafer and photolithographically patterned to define the metal pattern for the IDA circuit (Supporting Information, photomask design for the IDA devices.). A ~ 70 nm thick platinum film was sputtered onto the wafer, with a ~ 5 nm thick titanium layer to promote Pt adhesion to the substrate. The thicknesses were chosen to keep the metal layer at a sufficiently low sheet resistance while ensuring the sputtering process can be completed in a reasonable time and expense. Following sputtering, a liftoff process with Microposit 1165 solvent was performed to remove the photoresist. After liftoff, the wafer was plasma and RCA-1 cleaned again. Then, a ~ 1.5 μ m-thick SiO₂ layer was deposited on the wafer via plasma-enhanced chemical vapor deposition (PECVD). A KMPR 1010 photoresist was spun on and patterned by photolithography to expose the electrodes and the region for electrical connection. The exposed SiO₂ was then etched by a CF₄ plasma. After etching, a SPR220–4.5 photoresist layer was spun onto the wafer to protect the devices during the dicing process. The wafer was then diced into individual devices with a dicing saw using a resinoid blade (Advanced Dicing Technologies, Horsham, PA) (Figure S2, dicing procedure of the IDA devices.). The devices were degreased and O₂-plasma cleaned after dicing. Electrical connections between the flexible cables and the IDAs were made using anisotropic conductive tape, followed by pressing with a homemade thermal press to secure the connection. An image and an optical micrograph of the device is shown in Figure S3 (image and optical micrograph of the IDA device).

Electrochemical Measurements

All experiments were performed at room temperature. Electrochemical measurements were performed with a CHI760 bipotentiostat (CH Instruments, Bee Cave, TX), with the IDA placed in a 3D

printed electrochemical cell. For PPy, an Ag/AgCl/KCl (1M) (Ag/AgCl) reference electrode and either a graphite rod counter electrode (for electrode cleaning, validation, and PPy electrodeposition) or a 0.5 mm diameter Pt wire counter electrode (for ART measurements) were used. The reference electrode was put in contact with the solution via an agar gel salt bridge. For PTAP, all the electrochemical measurements were performed inside an Ar-filled glovebox. A PPy quasi-reference electrode (PPyQRE) was prepared according to a previously reported method and was calibrated using a Fc/Fc⁺ internal standard.⁵⁰ A graphite rod was used as the counter electrode. Individual electrode arrays on the microfabricated device were selected using a custom-built multiplexing circuit (Supporting Information, Arduino code to control the multiplexing circuit).^{42,43} All electrochemical measurements are reported in IUPAC convention.

Electrode Cleaning and Validation

Prior to performing ART measurements, the IDAs were electrochemically cycled in 0.1 M phosphate buffer solution (PBS, pH 7.4) to remove surface oxide formed during the oxidative plasma cleaning process (Figure S4, voltammetric behavior of cycling the IDA in PBS.). Then, to validate the performance of the IDAs, voltammetric generation-collection measurements were performed using either 1 mM FcMeOH with 0.1 M KNO₃ as the supporting electrolyte or 1 mM Fc with 0.1 M LiClO₄ as the supporting electrolyte (Figures S5–S6, generation-collection measurements for IDA device validation.).

Synthesis of 2,7-Dinitropyrene-4,5,9,10-tetraone Monomer

2,7-Dinitropyrene-4,5,9,10-tetraone (PTO-2NO₂) was synthesized according to previous reports with slight modification.⁵¹ A 45 mL, 4:1 v/v mixture of concentrated sulfuric and nitric acid (Sigma-Aldrich, MO) were charged into a flask with 844 mg (3.22 mmol) pyrene-4,5,9,10-tetrone (A2B Chem, San Diego, CA). The mixture was set to stir at 90 °C for 2 h before being poured onto 300 g of ice. The mixture was vacuum filtered to collect a bright orange precipitate that was washed with an additional 200 mL of water to yield the final product (959 mg, 85% yield). An ¹H nuclear magnetic resonance (NMR) spectrum of the final product is shown in Figure S7 (¹H NMR (400 MHz, DMSO-d₆) spectrum of 2,7-dinitropyrene-4,5,9,10-tetraone).

Electrodeposition

Before electrodeposition, the IDA electrodes were cycled in 0.5 M sulfuric acid to improve film adhesion (Figure S8, CV of IDA in 0.5 M sulfuric acid prior to PPy electrodeposition.). Electrodeposition of PPy was carried out using 10 mM pyrrole dissolved in MeCN, with 0.1 M TBAPF₆ as the supporting electrolyte. A CV was performed by sweeping the potential of the WE 1 array from the open circuit potential to 1.3 V vs Ag/AgCl under 500 mV/s scan rate, while the WE 2 array was held at 0 V vs Ag/AgCl. Electrodeposition of PTAP was carried out using 1 mM PTO-2NO₂ dissolved in MeCN, with 0.1 M LiClO₄ supporting electrolyte. A CV was conducted by sweeping the potential of the WE 1 array from the open circuit potential to -3.0 V vs Fc/Fc⁺ with 500 mV/s scan rate, while the WE 2 array was held at 0 V vs Fc/Fc⁺.

Redox Titration

ART measurements of PPy were carried out using 0.2 mM Ruhex as the redox mediator, with 0.1 or 0.5 M KCl as the supporting electrolyte. All solutions used in the ART measurements were purged with argon for 40 min and were kept under a humidified argon blanket before use. The electrochemical cell was tightly sealed with tape and was purged with argon for 10 min before solution was added. An argon filled syringe was used to add solution into the electrochemical cell, and the cell was immediately reclosed with tape after the solution being added. Argon flow was maintained throughout the ART measurements, and the flow rate was carefully adjusted to not cause convection of the solution. ART measurements of PTAP were performed using 0.5 mM Fc as the redox mediator with 0.1 M LiClO₄ supporting electrolyte. All measurements were carried out through automated Python scripts (Supporting Information,

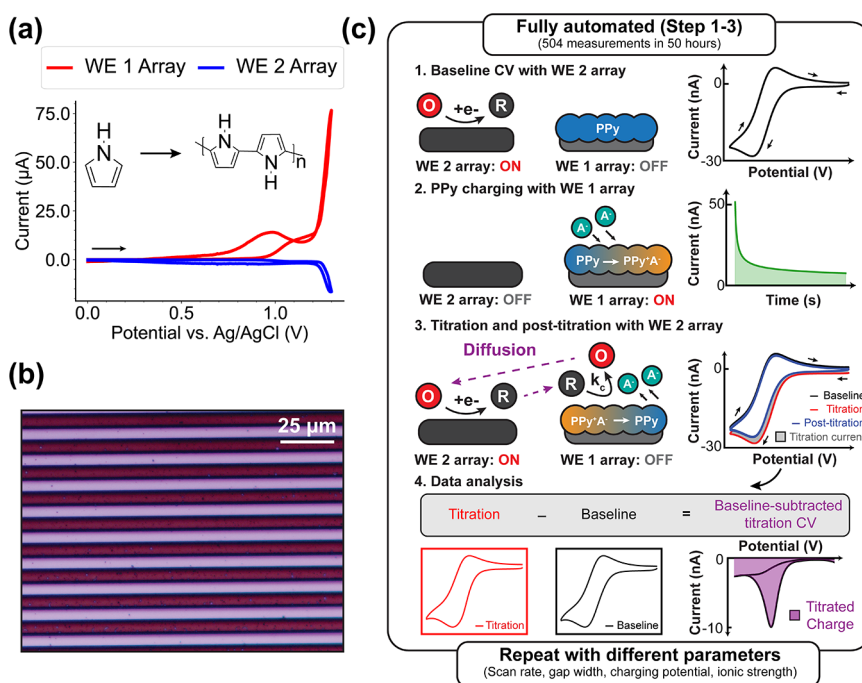


Figure 3. Electrodeposition of PPy and the automated workflow of the ART measurements. (a) Electrodeposition CV of PPy on the IDA with a 2 μm gap width. Inset schematic shows the chemical structure of pyrrole and PPy. (b) Optical micrograph (50-time magnification) of the 2 μm gap width IDA after electrodeposition. (c) The workflow of ART measurements (step 1–3) with the corresponding electrochemical response at each step. The amount of injected charge into PPy and the amount of titrated charge are represented by the shaded area of the electrochemical signal. The redox titration current and the amount of titrated charge were obtained by subtracting the baseline from the redox titration CV (step 4).

experimental Python scripts.) using Python libraries for hardware control developed by our group.^{9,52,53} An image of the entire experimental setup is shown in Figure S9, experimental setup for ART measurements.

Finite-Element Simulations

All finite-element simulations were carried out using COMSOL Multiphysics v5.6 with the transport of diluted species module (Supporting Information, COMSOL finite-element simulation models.). To save computational time, a simplified 2D IDA domain was constructed. The domain is 1 mm long along the y-axis and, along the x-axis, consists of one WE 2 electrode, two WE 2 electrode halves, and two WE 1 electrodes. All the electrodes were modeled as lines, with separations corresponding to the IDA gap width. Symmetry axes were applied at the boundaries of the x-axis, allowing the features in the domain to be replicated across these axes to generate the full IDA geometry. Boundary conditions and electrode reactions were incorporated accordingly, and a mesh that consists of the default number of elements in the bulk solution and finer free triangular mesh (50 elements) near the electrodes was built (Figure S10, overview of the simplified model for finite-element simulations.). The smallest mesh size was specified as ~ 30 nm, which is fine enough to model the flux on the edge of electrode surfaces, as simulations carried out with a less dense mesh offered noisy signal. (Figure S11, finite-element simulation results with a less dense mesh.) This allows the features near the IDA electrodes to be resolved more accurately while maintaining a reasonable simulation time. The out-of-plane thickness of the geometry was set to 510 μm , which is same as the length of the microband electrodes of the IDA. A detailed comparison between simplified and full geometry is shown in Figure S10 and S12 (simulation condition and result for the simplified and the full model.). Additional details about the finite-element simulations are available in Supplemental Notes S1, details on finite-element simulations.

Characterization of Materials

NMR spectra were collected using a 400 MHz NMR spectrometer (Bruker 400 MHz Avance III NMR with NanoBay console), equipped

with a SampleCase autosampler using IconNMR automation. Raman spectroscopy was performed using a Nanophoton Raman 11. Each spectrum was acquired using a 531.94 nm excitation wavelength, a LU Plan Fluor $\times 50/\text{NA } 0.8$ objective, and a 600 gr/mm grating with a center wavelength of 1420 cm^{-1} . The point measurements were acquired with a laser power of ~ 1 mW, while the XY Mapping was performed at ~ 0.1 mW. Each spectrum was acquired with an exposure time of 10 s and 5 averages.

RESULTS AND DISCUSSION

Electrodeposition of Polypyrrole

Following the fabrication and connection of the IDA devices, we first selectively electrodeposited PPy on the WE 1 array. The voltametric responses (Figure 3a, Figure S13, PPy electrodeposition CV.) show a large oxidative current on the WE 1 array, corresponding to the oxidative electro-polymerization of pyrrole.⁵⁴ On the other hand, a small reducing current was observed on the WE 2 array, which is due to the reduction of monomeric radicals generated from the oxidation of pyrrole monomers on the WE 1 array. Optical micrographs of an IDA after electrodeposition (Figure 3b) confirmed the selective electrodeposition of PPy on the WE 1 array, as the growth of a thin, purple-colored film exclusively on this array was observed, while the WE 2 array remained clean. The selectivity of electrodeposition was also confirmed by Raman spectroscopy (Figure S14, Raman spectrum for electrodeposited PPy on IDA.). We observed that biasing the WE 2 array to 0 V vs Ag/AgCl was crucial for the selectivity of the electrodeposition process, as it limits the diffusion layer of monomeric radicals to the region close to the WE 1 array (Figure S15, simulated behavior of PPy electrodeposition), resembling the principle of the electrochemical lensing process.⁵⁵ In contrast, if the WE 2 array was left unbiased, a diffusion layer of monomeric radicals spanning the entire IDA

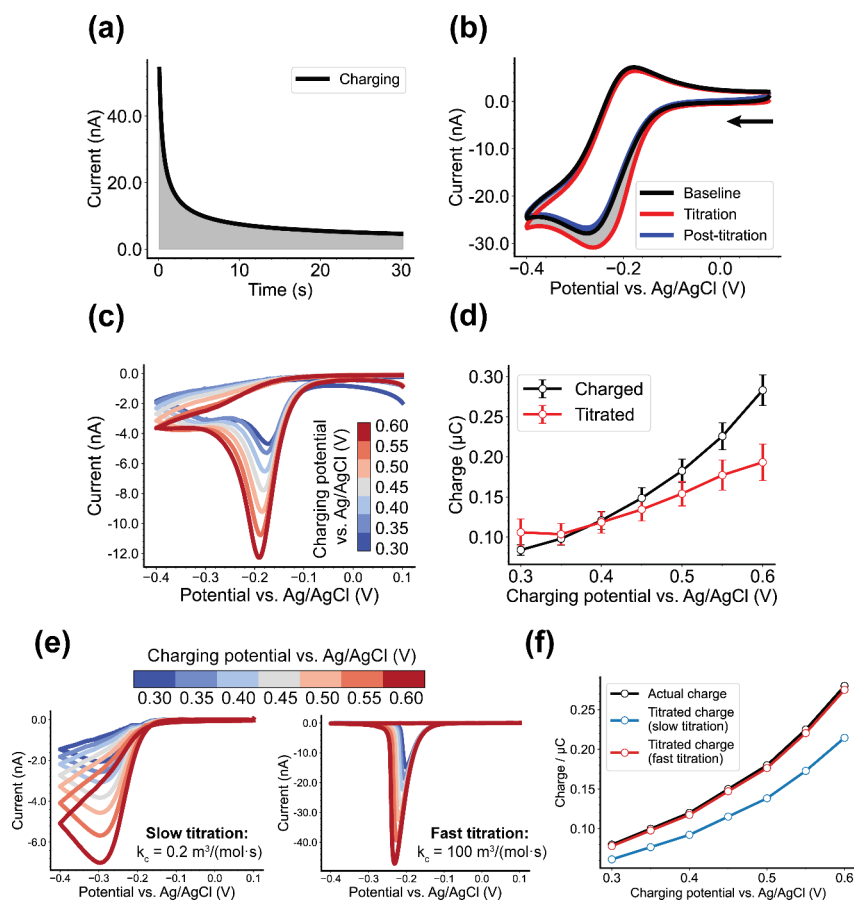


Figure 4. Electrochemical and simulated behavior of ART measurements on the 2 μm gap width IDA. (a) Charging curve of PPy at 0.6 V vs Ag/AgCl. The shaded area indicates the charge stored in the PPy thin film. (b) Overlay of the baseline, titration, and post-titration CVs following the charging curve shown in (a). The scan rate is 10 mV/s. The black arrow shows the initial scan direction of the CVs. The shaded area represents the difference between titration and baseline CVs. (c) Baseline-subtracted titration CVs plotted together. The color bar indicates the range of PPy charging potentials. (d) Stored charge and titrated charge plotted together versus the PPy charging potential. The error bars correspond to one standard deviation from three repeated measurements. (e) Simulated baseline-subtracted CVs at different PPy charging potentials for slow (left) and fast (right) titration kinetics. The color bar indicates the range of PPy charging potentials. (f) Comparison between the actual charge input into the simulation model and the titrated charge for slow and fast titration reaction kinetics at different charging potentials of the PPy thin film.

electrode area will form. As these monomeric radicals combine to form oligomers and eventually polymer chains (Figure S15), a precipitate will form on both WE arrays, compromising the selectivity of the electrodeposition.

Automated Redox Titration Workflow

The fully automated ART workflow for PPy is shown in Figure 3c. Once the solution is added to the electrochemical cell and all the experimental apparatus are properly connected, the only manual operation required is inputting parameters into a Python script (Supplemental Notes S2, details on the experimental Python script for studying PPy.), which controls the IDA and the electrochemical workstation to run experiments automatically.^{9,52} For the measurements, a baseline CV was performed on the WE 2 array. Then, the connection was switched to the WE 1 array, where an oxidative potential was applied for 30 s to charge PPy. Upon charging, PPy undergoes oxidation and incorporates ions from the electrolyte solution.⁵⁶ Finally, the WE 2 array was connected again to run CV with the same parameters as the baseline for three cycles, the first serving as the titration CV and the other two acting as the post-titration CVs. The completeness of redox titration is confirmed once the post-titration CVs overlap with the baseline.

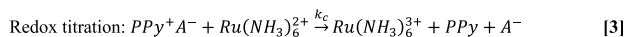
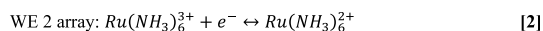
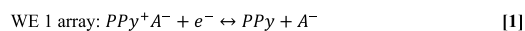
The programmability of our platform allows automated ART to be carried out with a large parameter space. We performed ART measurements on IDAs with different gap widths (2, 4, 6, and 8 μm) using different potentials to charge the PPy (from 0.3 to 0.6 V with 0.05 V increments, all referenced to Ag/AgCl) and running CVs with different scan rates (10, 25, and 50 mV/s) to explore the rate constant (k_c) of the titration reaction. With our automated platform, we completed a total number of 504 ART measurements within 50 h, under 2 supporting electrolyte concentrations. It took us less than 1 h of manual operations, mostly to clean, validate the IDA performance, and to program the experiment. Furthermore, studying different materials simply requires a different IDA device, highlighting the flexibility of our platform. The simplicity of the experimental setup allows researchers to explore a broad range of materials that would otherwise take unfeasibly long to study using the traditional SI-SECM setup.

Automated Redox Titration Measurements of Polypyrrole

We first performed ART measurements on electrodeposited PPy thin film using 0.2 mM Ruhex as a redox mediator in 0.1 M KCl supporting electrolyte. The data for the ART measurements on the 2 μm IDA at the various steps depicted in Figure 3, are shown in Figure 4 (data for other IDAs are

shown in Figures S16–17, all the PPy charging curves and baseline-subtracted CVs.). Figure 4a first shows the oxidative charging of the PPy thin film using a chronoamperometric charging curve at a potential of 0.6 V vs Ag/AgCl, thus prompting reaction 1 in Scheme 2 to form the oxidized form

Scheme 2. Electrochemical and Chemical Reactions Occurring during the ART Measurements of PPy



PPy⁺A[−]. Afterward, the WE 1 array was allowed to rest at open circuit, while the WE 2 array was activated to run a CV to generate the reduced form of Ruhex (Ru(NH₃)₆²⁺) according to reaction 2 in Scheme 2. Then, the titration CV, which prompted reaction 3 in Scheme 2, showed a current enhancement relative to the baseline (Figure 4b). The baseline in this case was obtained in the absence of intentional charging as described by reaction 1 in Scheme 2. We further verified the completeness of the reaction by a post-titration CV which overlaps with the baseline (Figure 4b), indicating that the charged PPy has been fully titrated.

To determine the charge recovery efficiency of our methodology, we first calculated the titrated charge by subtracting the baseline from all the titration CVs. A series of baseline-subtracted titration CVs is shown in Figure 4c. The magnitude of the current enhancement increases with the PPy charging potential, as observed through our charging curves and reported previously in the literature.^{57,58} We integrated the current of the baseline-subtracted titration CVs over the potential window and divided the result by the scan rate to calculate the titrated charge. We then integrated the current of the PPy charging curve with respect to time to calculate the charge stored within the PPy thin film (Supporting Information, sample Python script to perform charge integration.). A detailed description of the data analysis procedure can be found in Supplemental Notes S3, details on stored charge and titrated charge calculations. The titrated and stored charges at different PPy charging potentials are shown in Figure 4d.

The titrated charge shows a good agreement with the stored charge in the PPy thin film. However, discrepancies were observed at both high and low charging potentials. To understand the increased disagreement between the stored and titrated charges at higher PPy charging potentials, we applied finite-element simulations to model two cases of redox titration measurements: one with a slow titration reaction where the rate constant $k_c = 0.2 \text{ m}^3/(\text{mol}\cdot\text{s})$ and another case with a fast titration reaction where $k_c = 100 \text{ m}^3/(\text{mol}\cdot\text{s})$. The simulated baseline-subtracted titration CVs for the 2 μm IDA at 10 mV/s scan rate are shown in Figure 4e (simulated behavior for other interelectrode gap-widths and scan rates are shown in Figure S18, simulated redox titration behavior.). The simulation with fast titration reaction kinetics not only exhibits a higher current enhancement but also shows a different shape compared to the one with the slow titration reaction kinetics. This shape in the high kinetic regime, which in its original form before background subtraction is similar to that shown in Figure 2h, resembles the total catalysis scenario in molecular electrocatalysis theory.¹ The simulated CVs when slower titration kinetics are used show a much more gradual current

increase (Figure S19, simulated redox titration CVs). In the latter case, the current response was less sharp and shifted toward more negative potentials within the potential range of the mediator. The titrated charge plot in Figure 4f shows that there will be a consistent difference between the titrated and stored charge when the titration reaction is slow because the titration cannot be completed within the time scale of the CV measurement. Conversely, when the titration reaction is fast, no significant disagreement is observed, apart from the diffusional losses of the redox mediator caused by a collection efficiency less than unity. (Figure S5, generation-collection measurements with FcMeOH for IDA device validation.)⁴² The different simulated behavior clearly offered a qualitative understanding about the kinetics of the titration process. These finite-element simulation results enable our ART methodology to seamlessly distinguish redox titration kinetics that differ by 2–3 orders of magnitude. However, this is not the detection limit of our platform. With optimized experimental parameters, such as the CV scan rate, IDA gap width, and the amount of substrate material, we expect that our method can distinguish between MET systems with rate constants within the same order of magnitude with a reasonable signal-to-noise ratio. However, this will be approached in forthcoming works. These features hold close relevance to evaluating RTFB systems, as the MET kinetics is one of the most crucial design parameters and closely depends on the redox mediator-energy material combination. Being able to distinguish differences in MET kinetics offers our ART methodology the capability to optimize RTFB systems by screening different systems and obtaining their MET kinetics.^{31,34}

The simulations suggest that the kinetics of the titration reaction is the main factor that causes the experimental disagreement between the titrated and the stored charge in the PPy thin film. To this point, we speculate that the rate constant for titration could be a function of the amount of charge stored inside the polymer film. However, other factors such as overoxidation^{59–61} or structural degradation of PPy^{62–65} could also occur. The degradation of PPy has been confirmed by analyzing the amount of injected charge to PPy at fixed charging potentials under all three trials we performed, as we observed a decreased amount of injected charge into PPy throughout the ART measurements we performed (Figure S20, injected charge into PPy under each experimental trial.). With respect to the discrepancy at lower potential, there are additional factors which may contribute to an excess charge to that intentionally generated on the film. First, incipient oxidation via spontaneous charging of PPy at the open circuit potential^{65,66} may play a role. Despite our best efforts to remove O₂ from the solution, it is possible that a small portion of the film could spontaneously become oxidized to generate trace amounts of active species stable enough to be detected by redox titration. (Figure S21, CV responses illustrating the incipient charging of PPy.) Moreover, double-layer charging⁶⁷ results in a small amount of non-Faradaic current in the PPy charging curves. Since this background current are not easily removable from the charging curves, it causes a slight overestimation of the amount of injected charge. This effect will not affect the amount of titrated charge, because baseline subtraction during the evaluation of the titrated charge cancels out the non-Faradaic background in the baseline and titration CVs. Despite complications other than the redox titration kinetics, when the charging potentials are low, the amount of stored charge approaches the lower detection limit of the ART

method. However, a measurable titration response is still observed (Figure 4c), illustrating the sensitivity of ART at the low-signal limit.

One feature of ART is the ability to carry out systematic studies to explore how electrolyte conditioning may affect the titration process, as this is highly relevant to understanding processes in RTFBs. To elucidate the charge storage behavior of PPy under different supporting electrolyte concentrations, we performed ART measurements of the PPy thin film with 0.5 M KCl supporting electrolyte as well, while maintaining the concentration of the redox mediator constant. The electrochemical behavior on the 2 μm IDA at CV scan rates of 10 mV/s and 50 mV/s is shown in Figure 5 respectively. In

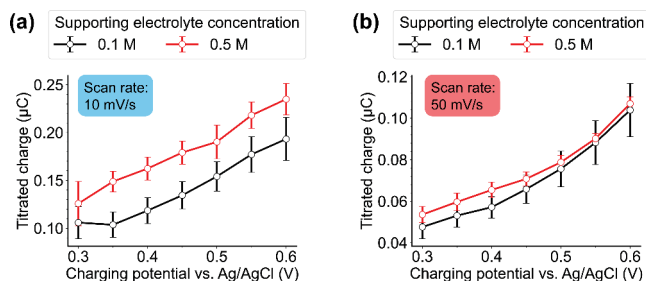


Figure 5. Electrochemical behavior of the PPy thin film under varied supporting electrolyte concentrations. Comparison of the titrated charge between 0.1 and 0.5 M KCl supporting electrolyte concentrations at CV scan rates of (a) 10 mV/s and (b) 50 mV/s.

general, the amount of charges stored in the PPy thin film increased under elevated electrolyte concentration because of the higher availability of anions in solution.⁶⁸ However, we observed that the titrated charge was only able to keep up with this trend at a low CV scan rate (10 mV/s). This observation is also attributed to the kinetics of the titration reaction. At a 10 mV/s scan rate, the titration reaction achieves the highest degree of completion within the time scale of the CV measurements and therefore allows most of the changes in the charge stored in the PPy thin film to be reflected by the titrated charge. Conversely, at higher scan rates, incomplete titration reactions cause them to not fully capture the changes of charges stored in the PPy thin film, and thus the titrated charge does not reflect the changes in the amount of stored charges.

In general, the slow kinetics of the titration reaction accounts for the lower amount of titrated charge compared to the amount of stored charge in PPy, as observed in the ART measurements compared to maximum collection efficiency obtained from generation-collection measurements with FcMeOH (Figure S5). Additionally, as the IDA gap width and CV scan rate increases, the charge recovery efficiency further decreases. (Figure S22, titrated and stored charge comparison for other CV scan rates and IDA gap widths.) These observations suggest that, depending on the system under investigation, not all the parameters may offer the desired sensitivity. Therefore, to fully understand a given system via ART using IDAs, researchers need to carefully design their experimental parameters, such as the CV scan rate, IDA gap width, concentration of the redox mediators, and the amount of substrate materials. These parameters highly depends on the system under investigation and should be iteratively optimized through the course of automated investigations. We also verified the stability of the electro-

deposited PPy thin film by examining the IDAs under an optical microscope after completing all the experiments. Images of the device revealed no noticeable delamination of the PPy thin film throughout the experiment. (Figure S23, optical micrographs of the IDA devices after experiment.) The ability to perform complex electrochemical redox titration measurements on the PPy thin film across a vast parameter space with programmability, minimal manual operation and supervision, and reproducibility highlights the capability of our ART platform to conduct systematic studies on a given material.

Automated Redox Titration of Pyrene-4,5,9,10-tetrone Azo Group-Based Polymer

Following the evaluation of PPy, we turned our attention to evaluating the charge storage behavior of PTAP, an emerging PTO-derivative which we believe could be used as a model system for RTFB studies. Importantly, we carried out these ART measurements inside an Ar-filled glovebox, highlighting the flexibility of our automated platform. We first electrodeposited PTAP on the WE 1 array of the IDA with 2 μm gap width. The electrodeposition CV shows several chemically irreversible reduction peaks on the WE 1 array, which correspond to the electroreduction of the nitro group to the azo group, as discussed by Li et al.⁴⁹ (Figure 6a-b). Formation of the azo group is crucial, as it serves as both the linker for the monomer units and redox-active site. Raman spectroscopy shows a peak at the Raman shift of the azo group (1424 cm^{-1}), as well as other characteristic peaks corresponding to PTAP, on the WE 1 array,⁴⁹ while WE 2 array displays negligible intensity. The Raman micrograph (Figure 6c) not only confirms the selective electrodeposition on WE 1 array, but also the integrity of the electrodeposited PTAP thin film, without significant pinholes and thickness variation.

We carried out ART measurements of PTAP using 0.5 mM Fc as the redox mediator, dissolved in MeCN with 0.1 M LiClO₄. The resulting CVs for the electropolymerized PTAP and the Fc redox mediator in the redox titration solution are shown in Figure 6d. The electrogenerated oxidized form of Fc, Fc⁺, will undergo a MET process with the reduced, lithiated form of PTAP (Figure 6d), generating a redox titration response analogous to that described previously for the PPy/Ruhex system, but now encoded in the redox behavior of Fc. Following a workflow similar to ART measurement of PPy (Figure S24, ART workflow for studying PTAP. Supplemental Notes S4, details on the experimental Python script for studying PTAP.), we first lithiated PTAP with increasing negative-going charging potentials and performed redox titration of lithiated PTAP with electrogenerated Fc⁺ afterwards. The baseline-subtracted titration CVs using the Fc/Fc⁺ system at different charging potentials for PTAP are shown in Figure 6e, carried out using the IDA with a 2 μm gap width. At lower charging potentials, the carbonyl groups of PTAP become lithiated, and a peak shows up on the baseline-subtracted titration CVs, corresponding to the redox titration of lithiated carbonyl groups by Fc⁺. However, as the charging potential of PTAP becomes more negative, there is a shift in the redox titration response, where there is a gradual displacement toward more positive potentials and then a split into two peaks. These two observations suggest a decrease in the rate of the MET process as the charging potential of PTAP gets larger. The appearance of two peaks has been ascribed in SI-SECM literature to the presence of sequential

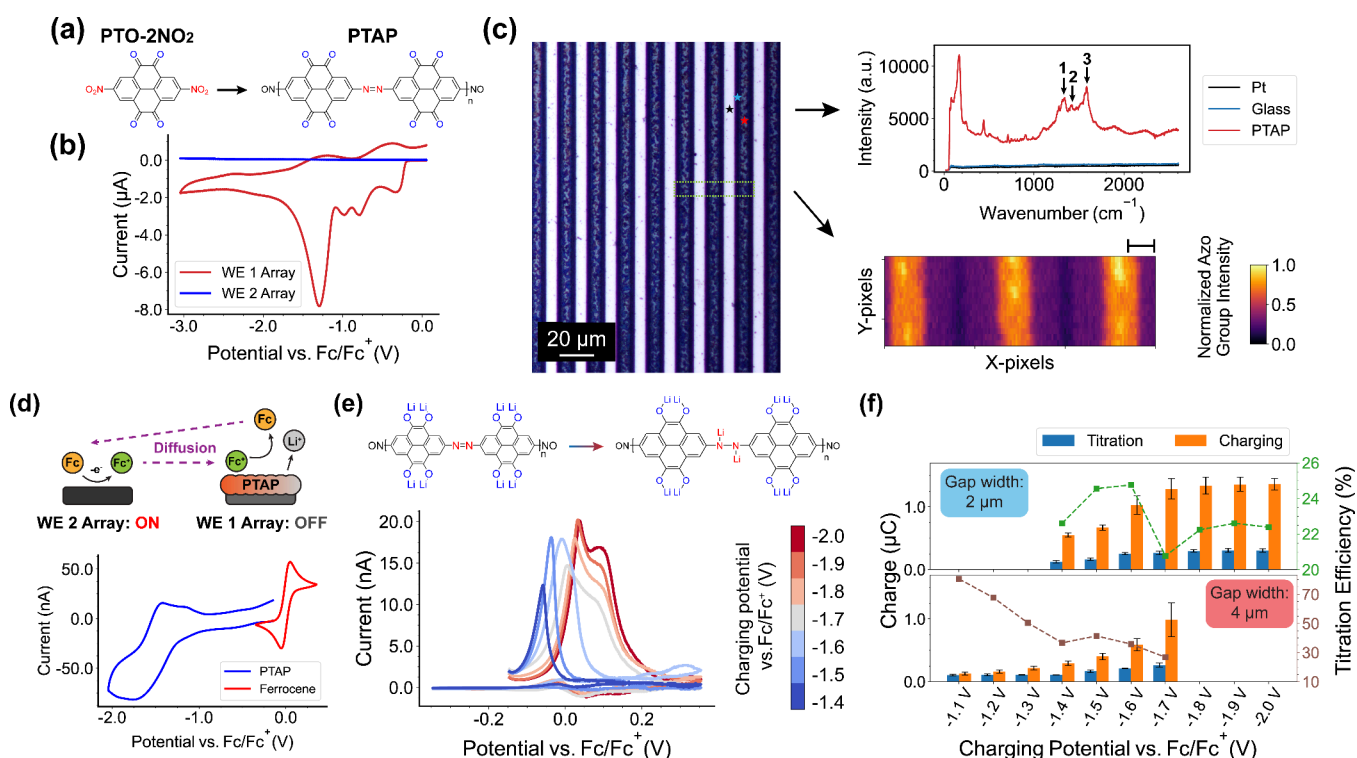


Figure 6. ART measurement results of PTAP. (a) Chemical structure of PTO-2NO₂ monomer and PTAP. (b) Electrodeposition CV of PTAP on the IDA with 2 μm gap width. (c) (Left) Optical micrograph of the IDA after the electropolymerization of PTAP, with the region where Raman spectroscopy was taken labeled. (Top right) Point Raman spectra of glass, clean Pt electrode, and PTAP deposited region on the IDA. The points where the spectra were taken are labeled on the micrograph. The labeled peaks on the spectra are characteristic peaks for PTAP. Their assignments are as follows. 1: Nitro group (1341 cm⁻¹); 2: Aromatic azo group (1424 cm⁻¹); 3: Aromatic C = C (1585 cm⁻¹). (Bottom right) Raman micrograph of the scanned region. The intensity shown in the micrograph corresponds to the wavenumber for the azo group, 1424 cm⁻¹. The scale bar is 5 μm. (d) CV of PTAP and Fc taken after the electrodeposition, in a solution of 0.5 mM ferrocene, 0.1 M LiClO₄, dissolved in MeCN. The scan rates for PTAP and Fc are 50 mV/s and 100 mV/s, respectively. The current of the Fc CV is scaled to 1/5 of its original value for the clarity of the illustration. (e) Baseline-subtracted titration CVs at all the PTAP charging potentials, obtained from the IDA with 2 μm gap width. The chemical structures of PTAP when (left) charged at lower potential and (right) charged at higher potential are also shown. (f) The comparison between the amount of injected charge during PTAP charging and the amount of titrated charge. The titration efficiency, which is the ratio between the titrated and injected charge, is shown on the secondary y-axis. (Top) Data obtained from the IDA with 2 μm gap width, using higher PTAP charging potentials. (Bottom) Data obtained from the IDA with 4 μm gap width, using lower PTAP charging potentials.

electron transfer events, which occur when multiple titratable species - in parallel or in sequence- can form in a system.³⁹ Since the azo groups in PTAP become lithiated at higher charging potentials, we associate this additional peak on the baseline-subtracted redox titration CVs with the redox titration of lithiated azo groups. Regarding the potential-dependent redox titration behavior, it is important to notice that the observed shifts do not occur due to changes in the position of the mediator CV (Figure S25, additional electrochemical results for PTAP.). We speculate that changes in MET rate could be due to differences in the microenvironment offered by the PTAP polymer as a function of the charging potential - a phenomenon observed broadly across redox-active polymers,⁶⁹ or it could be signaling more subtle effects in MET dynamics, such as a decrease in rate constant for the MET process as a function of the energy difference between polymer and mediator, i.e., a Marcus inverted region-type phenomenon⁷⁰ and functional group-dependent redox titration kinetics. Further studies, such as redox titration with alternative redox mediators, are needed to elucidate all the causes, but clearly our ART methodology creates the opportunity to systematically detect these reactive trends.

We finally turn to the charging-titration behavior of PTAP (Figure 6f). We observed a sudden increase in both the

injected and the titrated charge at -1.7 V vs Fc/Fc⁺ charging potential, which aligns with the appearance of the additional peak at the most negative charging potentials in the ART results in Figure 6e. This observation supports the possible onset of azo group lithiation at -1.7 V vs Fc/Fc⁺. However, for all these experiments, the ratio of titrated to injected charge, i.e., the titration efficiency, showed values of less than 25%, as shown in Figure 6f. This evidence chemical irreversibility that could be due to different charging potentials and their associated chemical intermediates. To this point, we used an IDA with 4 μm gap width to explore the behavior at lower charging potentials. Indeed, although the charge stored is significantly smaller, charging PTAP between -1.1 and -1.3 V vs Fc/Fc⁺ evidenced a much-improved titration efficiency of up to ~ 80%, with an observable drop after -1.4 V vs Fc/Fc⁺ (Figure S26, experimental results with lower PTAP charging potentials.). We highlight that determining these reactive trends is nontrivial from the analysis of the CV in Figure 6d, which only shows a shoulder peak at these low charging potentials. Another factor that could lead to the low titration efficiency obtained is the Li⁺ diffusion in the electrolyte. However, based on the calculated Li⁺ diffusion coefficient in MeCN ($D_{Li^+} = 4 \times 10^{-6}$ cm²/s),⁷¹ the characteristic diffusion time ($t = d^2/D_{Li^+}$, where $d = 2-4$ μm) of Li⁺ from the WE 1

to the WE 2 array is 10 – 40 ms, much less significant than the time scale of the ART measurements. Moreover, the use of concentrated supporting electrolyte (0.1 M) will minimize the solution resistance for ionic transport. Therefore, Li^+ transport in electrolyte is probably not the primary reason for the low titration efficiency obtained. Other possible tests for the observed irreversibility such as modifying the charging time to explore the potential impact of Li^+ diffusion into the PTAP film, did not result in observable differences (Figure S27, experimental results with shorter PTAP charging duration). Thus, improving the overall redox targeting behavior of a material such as PTAP might rely on identifying the intermediates and potential regions leading to higher rates and efficiencies, which our new ART methodology promptly elucidates. Our IDA methodology also shows highly reproducible results throughout different microfabricated samples, as data shown in Figure 6f were obtained from IDAs on different samples, and a highly reproducible titrated charge versus charging potential trend was identified Figure S28 (titrated charge reproducibility throughout different microfabricated devices).

CONCLUSIONS

In this work, we introduced interdigitated electrode arrays (IDAs) as a platform for automated redox titration (ART) measurements, enabling efficient study of the mediated electron transfer (MET) process for energy storage materials of interest in redox targeting flow batteries (RTFBs). Finite-element simulations were used to illustrate the characteristic voltammetric responses of ART measurements and provided insights into the MET kinetics, which are crucial for RTFB dynamics, and as demonstrated here, of high relevance when exploring the titration of redox-active polymers.

We demonstrated the utility of our methodology by investigating two redox-active polymers, polypyrrole (PPy) and pyrene-4,5,9,10-tetrone azo group-based polymer (PTAP), and revealed their charge storage behavior under varied conditions. PPy was used first as a model system, exhibiting a high titration efficiency, i.e., the ratio of electrode-injected to mediator-titrated charge, and few complications from the kinetics of the MET process. The sensitivity of this system to factors such as electrolyte ionic strength was assessed to highlight the capability of conducting systematic studies at high throughput using the ART method. On the other hand, electropolymerized PTAP, a redox polymer of interest for emerging RTFBs, showed a strong dependence on MET kinetics and titration efficiency as a function of the potential limit to which it was charged, displaying a marked decrease in efficiency from ~ 80% at the least negative charging potentials down to ~ 25% at higher charging potentials. These changes were also accompanied by an unexpected decrease in the rate of MET as a function of charging potential, highlighting the importance of developing new tools such as ART that can capture these kinetic effects which may decisively impact the efficiency of the redox targeting process.

Our platform incorporates microfabricated IDA devices with four different electrode gap widths, a multiplexing circuit, and a Python library for hardware control, and, as demonstrated here, it can be set up in the air-free environment of a drybox. This methodology dramatically simplifies the analysis done with SI-SECM techniques, offering a streamlined alternative to SECM approaches by introducing programmable automation, a capability not previously established. Our platform enables

the minimization of human error, expansion of studies to a larger library of materials, and close-loop studies when coupled with machine-learning algorithms.⁷² Our next direction is to apply our platform to high-throughput screening of next-generation energy storage materials,^{9,24,29,35,73} particularly for applications in RTFBs. Here, we established the possibility of using electropolymerization as a convenient means not only to study materials, but with potential applications for creating monolithic structures for use in RTFBs. Planned integrations with automated solution handling platforms^{9,43} will enable systematic studies of other air-sensitive and practical materials. We also aim to extend the platform's compatibility to nonelectrodeposited materials - such as catalysts, redox-active colloids, and self-assembled monolayers-via new surface-patterning strategies. We look forward to expanding the capabilities of our integrated ART and automated electrochemistry platform to address more comprehensive topics in the field of electrochemistry.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsmeasuresciau.5c00128>.

Experimental Python scripts (IPYNB), Sample Python script to perform charge integration (IPYNB), Arduino code to control the multiplexing circuit (INO), COMSOL finite-element simulation models (MPH), and photomask design for the IDA devices (GDSII) (ZIP)

All the experimental data files (ZIP)

Details on device fabrication and packaging, finite-element simulations, experimental Python script, and data analysis. Additional electrochemical measurements and finite-element simulations results (PDF)

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Notes

The authors declare no competing financial interest.

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